Results: Language, Topics, and Embeddings

RQ3: Topical and Linguistic Structure I

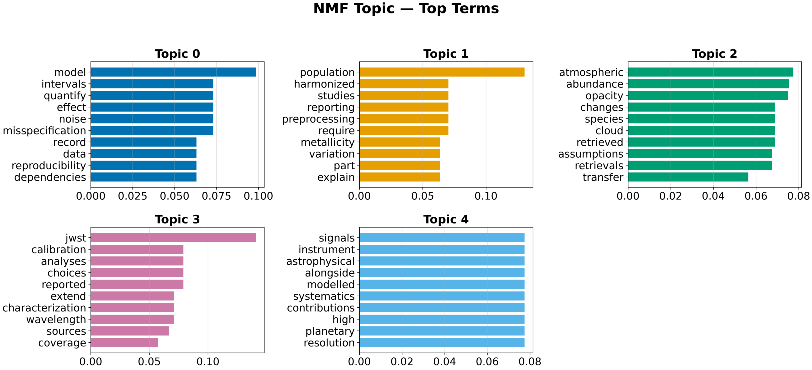
Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 137-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: atmospheric, uncertainty, spectra, temperature, abundance, coverage, opacity, sources, analysis, assumptions, observed, retrievals, models, stellar, across, evaluated, retrieval, structure, composition, jwst.

Table 3. NMF topics extracted from the corpus.

Topic	Top terms
0	model, intervals, quantify, effect, noise, misspecification, record, data
1	population, harmonized, studies, reporting, preprocessing, require, metallicity, variation
2	atmospheric, abundance, opacity, changes, species, cloud, retrieved, assumptions
3	jwst, calibration, analyses, choices, reported, extend, characterization, wavelength
4	signals, instrument, astrophysical, alongside, modelled, systematics, contributions, high

RQ3: Topical and Linguistic Structure II

The topic labels are computed from the retained corpus and are not hand-assigned. Their interpretation should be checked against the generated topic-term weights and the configured subfield taxonomy; fixture-derived topics describe synthetic text patterns, not the scientific field.



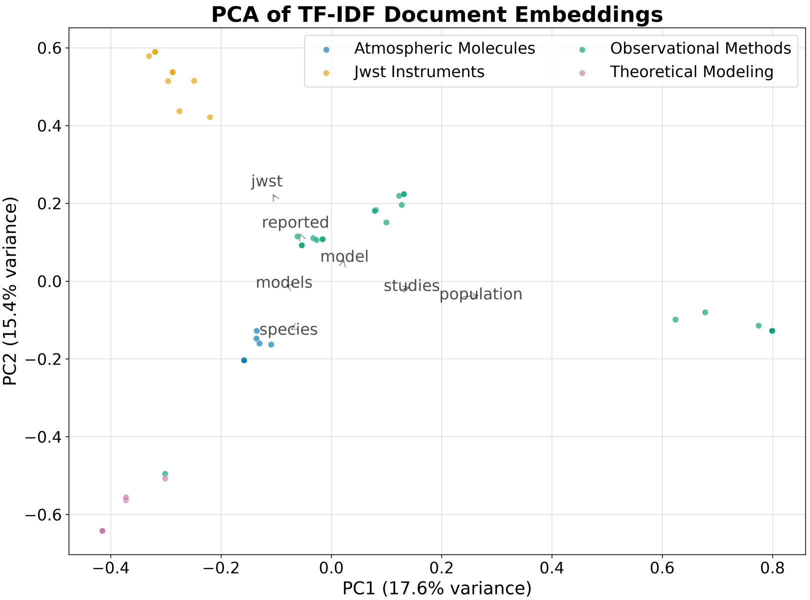
RQ3: Topical and Linguistic Structure III

Figure 1: Topic-term bar charts for Exoplanet Atmospheres. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the $\mathbf{H}$ matrix.

Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

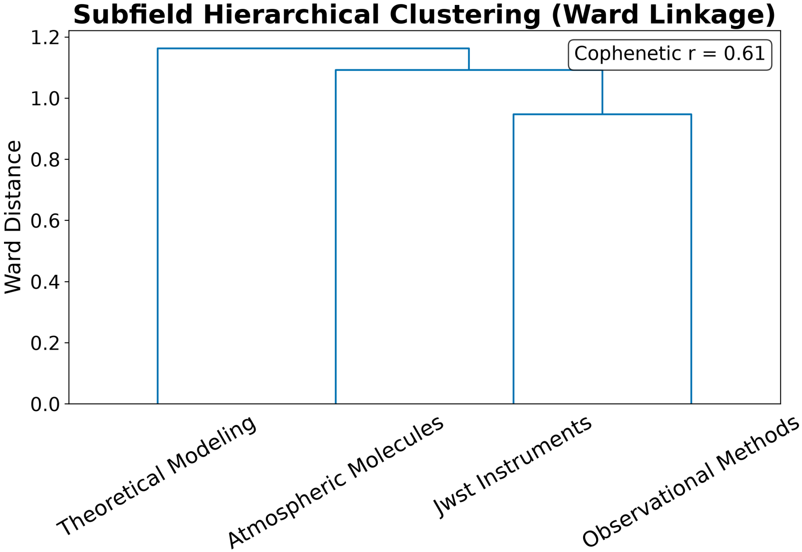
Document Embeddings II



Document Embeddings III

Figure 2: PCA projection of document embeddings for Exoplanet Atmospheres. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

Document Embeddings IV

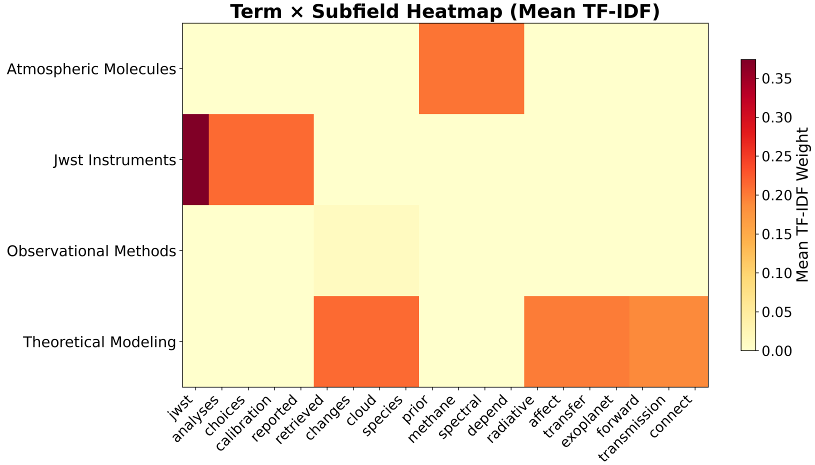


Document Embeddings V

Figure 3: Hierarchical clustering dendrogram of document embeddings. The tree shows the similarity structure of the corpus: documents that join low in the tree are semantically similar, while high-level splits separate the major topical clusters.

Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.



Term Analysis II

Figure 4: Term heatmap for Exoplanet Atmospheres. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

Named Entity Analysis I

Named entity extraction over the 46 abstracts identified 1 unique entities. The most frequent entities reflect the retained corpus and the configured extraction rules; source coverage and fixture status determine how they may be interpreted.

Table 4. Top named entities in abstracts.

Entity	Frequency
JWST	15

Named Entity Analysis II

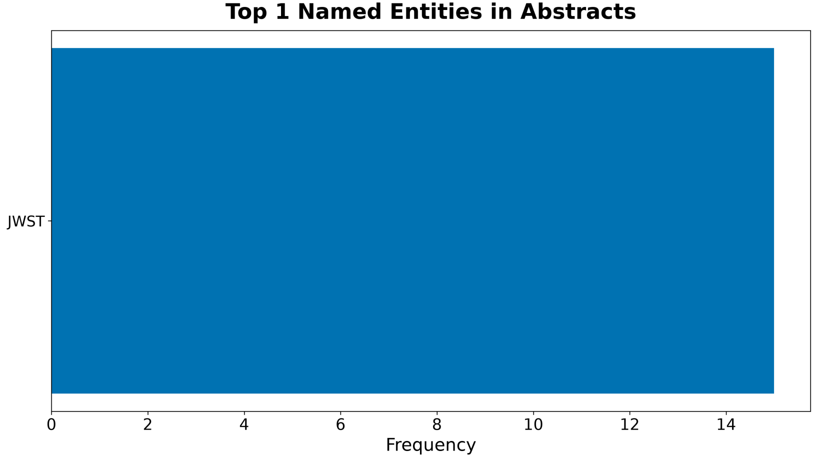


Figure 5: Top named entities for Exoplanet Atmospheres. The horizontal bar chart shows the 20 most frequently extracted entities from abstracts, revealing recurring objects, instruments, methods, and concepts in the retained corpus.

Named Entity Analysis III

Named Entity Analysis IV

Table 5. Top keyphrases by TF-IDF score.

Keyphrase	Score
jwst	0.0714
population	0.0556
analyses	0.0435
calibration	0.0435
calibration choices	0.0435
choices	0.0435
combined	0.0435
jwst analyses	0.0435
models	0.0435
models calibration	0.0435
models calibration choices	0.0435
model	0.0400
atmospheric	0.0392
abundance	0.0385
assumptions	0.0385

Embedding Similarity and Clustering I

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

Table 6. Top 10 most similar document pairs.

Paper A	Paper B	Similarity
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000
doi:10.5555/exoplanet atmosphe	doi:10.5555/exoplanet atmosphe	1.0000

Embedding Similarity and Clustering II

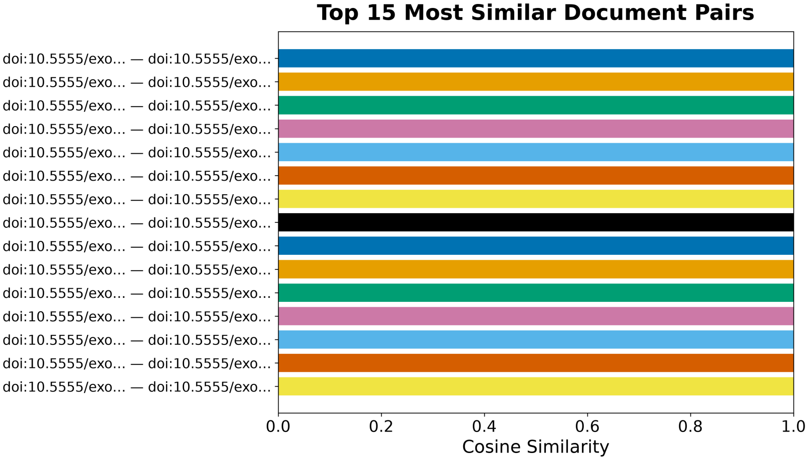
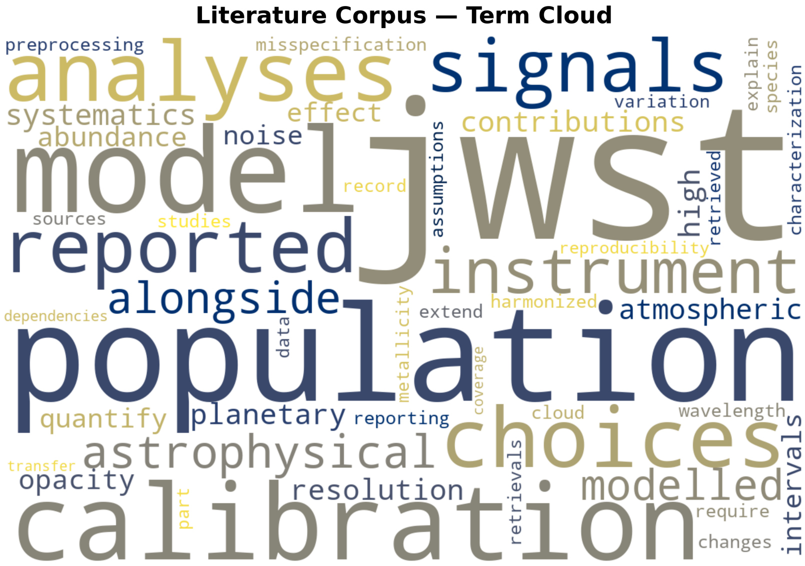


Figure 6: Document similarity for Exoplanet Atmospheres. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

Embedding Similarity and Clustering III



Embedding Similarity and Clustering IV

Figure 7: Term cloud for Exoplanet Atmospheres. Term sizes are proportional to their TF-IDF weights across the corpus, providing a visual summary of the dominant vocabulary.

Embedding Similarity and Clustering V